P515 /1

WAKISSHA JOINT MOCK EXAMINATION

PRINCIPLES AND PRACTICES OF AGRICULTURE

PAPER 1

(Theory)

3 HOURS

STUDENT-BASED GUIDE TO WAKISSHA MOCK AGRICULTURE PAPER 1

NOTE;

The above Marking guide for the Agriculture Mock Examination Paper 1 has been revised and attempted by an Agriculture student for the UACE 2026 curriculum.

The answers and marking points have been developed in line with the new competency-based curriculum, with emphasis on scenario-based questions, Practical agricultural knowledge, application of concepts, and appropriate vocational agriculture approaches.

Where necessary, relevant agricultural research and reperence Tools and Materials have been considered to ensure that the Marking guide is accurate, reliable and suitable for assessment.

IMPORTANT NOTICE;

Teachers and students are requested to read the marking guide carefully and make any necessary adjustments, additions, or corrections where appropriate.

Any suggested corrections should be based on sound agricultural knowledge, the requirements of the curriculum, and accepted agricultural practices.

Feedback, corrections, and professional contributions are highly welcome.

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For further clarification, interpretation, and scruting of the Agriculture Marking guide. Please contact;

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Thank You for your cooperation and contribution towards improving the quality of Agriculture assessment.

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ITEM · 1

Part (a)

Production far below expected milk yield despite good management; carry out a full review of feeding, breeding and health management as this helps in identifying and correcting of the specific gaps such as genetics, nutrition or health inorder to raise milk yield towards its potential thus resulting into increased income from milk sales

Milk kept in dirty containers, usually left open in a living room; Adopt use of clean, well-sealed containers stored in a clean, cool dedicated place as this prevents bacterial contamination and off-flavours hence preserving milk quality and preventing low prices from poor hygiene.

Milk stored in a metallic container made of copper; Mr. Kakeeto should switch to use of stainless steel containers because copper reacts with milk components and catalyses oxidation of milk fat, causing off-flavours and destroying nutrients which lowers milk quality

Obtained an agro-loan with a high interest rate of 35%; seek loans from other financial institutions with a low interest rate, this will ensure more of the returns remain as profits rather than going to loan repaymen t.

Selling only raw milk despite good milk quality; Add value to meat products by processing them into cheese, yoghurt, butter, ghee or ice cream because value-added products attract more customers, last longer and earn higher profits

No clear market strategy for higher-value products; Advertise and promote milk products through radio, television, newspaper, posters and social media platforms, this creates awareness about the products attracting new customers and encouraging the existing customers to buy more.

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ITEM - 1

Part (b)

CHEESE

This is a milk product made by coagulating milk using rennin obtained from calves or synthetically made.

It can be used for cooking or grating over salads

MATERIALS AND EQUIPMENTS

Fresh milk	stirrer	Protective gear
Rennet enzyme	Knife	starter's culture
Salt	Heat sauce	Packaging material
Sauce Pan	Muslin cloth	labels.

PROCEDURES

Put on protective clothing such as hair net, gloves and apron for body safety and hygiene standards.

Assemble all materials and equipments for use in making cheese to ensure smooth conduct of the project.

Wash the saucepan and knife with clean water to avoid contamination of the milk.

Pasteurize fresh milk upto 78°C for 20 minutes to kill harmful bacteria and cool to make milk hospitable for good bacteria.

Add starter culture into the milk to introduce bacteria that converts lactose into lactic acid

Add rennet enzyme or acid to coagulate milk forming curds and whey.

cut, stir, cook and wash curds to release whey.

Drain the curds using a sieve or muslin cloth to remove moisture.

Add salt and age the cheese to add good taste and flavour.

Cut the cheese into desired form or shape before use to ease packaging.

Package cheese in materials and label with ingredients, health benefits, logo, production date and business contact.

Sell through local shops, schools, markets, restaurants, hotels or use online platforms such as whatsApp, Tiktok and Youtube.

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ITEM -2

Part (a)

Persistent wilting of tomato plants during the day even though the soils appear moist; The roots are unable to absorb water fast enought to replace what is lost through transpiration, therefore cells lose turgor pressure causing leaves and stems to drop thus reducing photosynthesis which lowers plant growth and fruit yield.

Yellowing of leaves with scorched edges; Yellowing indicates lack of chlorophyll molecules from nutrient deficiency such as nitrogen, reducing the leaves' ability to trap sunlight for photosynthesis and manufacture food. leading to stunted growth and poor fruit formation.

leaf cells observed under microscope are flaccid with reduced internal volume and poorly defined internal structures; Flaccid cells confirm low turgor pressure from water shortage, while poorly defined internal structures show impaired photosynthesis and respiration. This causes cells to produce low ATP leaving the plants unable to sustain normal growth and fruit formation.

Overall plant growth remains stunted and fruit formation is poor despite continued fertilizer use; Since the roots cannot effectively absorb nutrients due to their damaged state, the fertilizer applied goes to waste rather than reaching the plant tissue thus plants therefore remain under nourished, stays stunted, and produces few, poorly developed fruits, directly reducing the harvest.

Dairy cattle show reduced milk despite being fed regularly; indicates animals digestive system is not properly breaking down and absorbing nutrients from the feed, since milk synthesis requires a steady supply of glucose, amino acids and fatty acids delivered to the udder, therefore poor absorption means the mammary glands lack these raw materials, so milk yield falls

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2 part. a

Gradual weight loss and general body weakness in cattle; Because the animals cannot absorb properly nutrients from their feeds, their bodies break down their stored fats and eventually body proteins for energy causing weight loss and weakness resulting into under weight animals that cannot sustain high milk production

Poorly developed muscles; Mobilization of proteins and fats for energy production reduces protein percentage in the body, this interferes with body tissue formation resulting into poorly developed muscles in the udder thus reduction in milk production.

Fatigue after minimal movement; Muscle contraction requires sufficient ATP supply from aerobic respiration, reduction in ATP supply causes these animals cells to depend on anaerobic respiration for ATP supply which also yield lactic acid which accumulates limiting movements to grazing areas for feed intake thus leading to slow growth and reduced milk.

Reduced apetite; This may be due to digestive disorders which interferes with normal appetite for feeds, reducing feed intake and causing a decrease in energy and nutrient supply thus worsening weight loss and poor production

ITEM - 2

Part (b)

Continuous use of fertilizers has reduced soil productivity; carryout soil testing and apply fertilizers according to the nutrients found to be depicient incor porating with well-decomposed organic manure. this helps to identify the nutrients lacking in the soil while organic manure adds nutrients and improves soil structu re and water holding capacity hence promoting better root development and nutrient uptake resulting into vigorous tomato growth

Tomato plants showed poor cell turgor and photosynthetic efficiency in leaves; Ensure consistent, adequate irrigation and mulch to conserve soil moisture and improve soil fertility, this maintains steady water availability to keep cells turgid and metabolically active thus supporting normal growth and fruit formation.

Tomato wilting despite moist soil; Carryout deep ploughing to break down the soil solids to ensure soil drainage, this ensures soil aera tion allowing roots to respire aerobically to produce energy needed for active uptake of nutrients which facilitates osmosis in return Hence maintaining the plant cell turgor for improved photosynthesis.

Leaf yellowing and scorching : conduct a soil test to identify the specific depiciences and apply the correct fertilizer for example NPK fertilizer that can supply nitrogen needed for restoring chlorophyll productio n, this enables normal photosynthetic process to take place. resulting into high yields

Poor fruit formation despite fertilizer use; Address the root health issue first before relying on fertilizer, and apply fertilizer in split does matched to plant uptake capacity. This ensures nutrients applied are actually absorbed and used for fruit development rather than being wasted.

Low milk yield ; Provide a balanced diet containing adequate energy proteins, minerals, vitamins and fibre according to the animals age and production stage, as this provides ATP for cellular activities, while

proteins provide amino acids for muscle and tissue formation therefore improving body condition, growth and milk production.

Cattle have poorly developed muscles and slow growth ; should include suitable protein-rich feeds such as legumes and other appropriate protein sources in the ration since proteins are digested into amino acids which are absorbed and used to synthesize muscle proteins, enzymes, hormones and other tissues, therefore promoting muscle development, growth and repair.

Cattle lose weight, become weak and produce less milk ; should provide adequate energy-rich feeds while maintaing sufficient roughage for proper rumen function. Energy is required to produce ATP through cellular respiration thus adequate energy prevents excessive mobilization of body fat and protein and allows nutrients to be directed towards growth and milk synthesis, increasing productivity.

Animals show weakness, weight loss, rough coats and poor growth ; should have the animals examined by a qualified veterinarian and follow the recommended treatment and management programme since the symptoms may arise from diseases, parasites of the digestive system. Identifying and treating the underlying causes restores normal physiological functions and prevents further losses.

Poor environmental conditions on the housing and structures ; should then maintain clean, well-ventilated and comfortable cattle housing and provide waste disposal, good hygiene therefore reduces exposure to pathogens, while adequate ventilation reduces respiratory stress hence resulting into better growth and milk yield since healthy animals use more of their nutrients for production rather than immune responses.

Poor appetite and reduced productivity ; should establish an appropriate parasite-control programme under veterinary guidance since parasites compete for nutrients, damage tissues and may cause blood loss therefore removing them allows more nutrients to be absorbed and utilized by the animal for growth, maintenance and milk production.

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ITEM · 3

Part (a)

Animals struggle for water around the watering points; Insufficient water supply at water points means animals don't drink enough water and since water is essential for milk synthesis, the low intake directly lowers the milk secretion by the udder leading to low yield

Kraal has uneven, rough finish floor; The floor injures the animals hooves and legs causing pain and discomfort that reduces movement of animals to feeds and water, leading to low feed intake and reduced milk production.

Animals move to nearby areas with other neighbourhood animals during dry season; Mixing animals with other herds exposes animals to diseases and parasites and animals may easily get lost reducing the herd size resulting into reduced amount of milk production on the farm.

Cattle graze on young fresh lush grass immediately after raining; lush grass ferments rapidly in the rumen producing excess gases that cannot be released fast enough, causing froth bloat. This discomfort reduces feed intake and reducing nutrients in the animal body leading to reduced milk production.

Feeding schedule varies through the week; Milk synthesis needs a steady supply of nutrients, reducing food and water towards the weekend starves the udder of these requirements. Thus leading to a drop in milk production from 15 to 6 litres.

Calves suckle continuously; Continuous suckling stimulates the calf's oxytocin-driven milk let-down reflex that empties the udder before milking thus leaving less milk for the farmer

Bulls graze freely with cows; This may lead to uncontrolled mating of bulls with the heifers causing the risks of inbreeding giving rise to the off springs with genetic vigour potential that is very poor, leading to less production of milk.

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3

part a

Blood and clots observed in milk; Indicates mastitis which damages the milk producing tissues of the udder and contaminates the milk, causing reduction in both the quantity and quality of milk obtained thus fetching low profits to the farmer.

Animals are milked at any time of the day; Milk let-down depends on the hormone oxytocin which is triggered by regular milking routine therefore irregular milking times disrupts this reflex making milk to not be fully released from the udder, hence lowering the amount of milk collected.

Milk is contaminated with fur and other foreign materials; Poor milking hygiene introduces bacteria and dirt into the milk, causing it to spoil quickly and become unfit for sale, reducing the farmer's marketable milk out put and income.

Wrong acaricide dilution of 30L for 25 cows; under-dosing leaves the ticks under-exposed to the acaricide active ingredients, allowing survival and even ~~survival~~ resistance development enabling ticks to suckle blood and transmit diseases that destroy blood cells causing anaemia and fever that suppresses apetite and milk production.

ITEM · 3

Part (b)

Animals struggle for water; Provide adequate clean water troughs around the grazing yard, this ensures sufficient water for digestion, temperature regulation and milk production thereby increasing productivity

Animals move away during the dry season looking for pasture; He should establish pasture plots and fodder banks and conserve excess grass as hay or silage during the rainy season, conserved fodder provides feed when natural pastures become scarce · This prevents animals from losing body weight and reduces competition with neighbours animals

Feeding varies greatly between days; Kakamega should provide a consistent balanced ration based on age, body condition and production levels while supplementing pastures with hay, silage or concentrates · consistent feeding provides adequate energy and nutrients for maintainance, growth and milk production, thus improving body condition and prevents fluctuations in production.

Cattle are grazed on young, fresh lush grass after rainfall; He should practice rotational grazing, allowing some paddocks to rest and recover as it prevents continuous removal of young shoots and allows grass to regenerate providing a sustainable supply of nutritious forage, maintaining animal productivity over a longer Period.

Bulls freely mix with adult female cattle; Separate bulls from females and introduce them only during planned breeding periods since good breeding management can produce offspring with desirable productive characteristics, hence improving the herds future milk and growth performance.

Calves suckle their mothers continuously; Practise controlled suckling and timely weaning, while ensuring calves receive adequate nutrition · This ensures calves receive enough milk for growth without unnecessarily consuming all the milk produced by the cow hence more milk can be collected for sale, while proper calf feeding maintains healthy growth.

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3

Part - b

Blood and clots are present in milk; should examine and treat affected cows by a veterinary professional, and maintain proper udder hygiene. Treatment controls the infection and helps restore normal udder function as this improves both milk quality and quantity.

Milking is done at different times of the day; Milk the cows at fixed times ~~of~~ every day, maintaining approximately equal intervals between the milking, because regular milking establishes a consistent physiological routine and helps maintain efficient milk secretion and let-down. This prevent to unnecessary changes in milk yield and improves amount of milk obtained.

Milk is ~~contaminted~~ contaminated with fur and foreign materials; clean the udder, hands and milking equipment before milking, keep the milking area clean and filter after milking since clean milking practices prevent dirt, hair and micro organisms from entering the milk, this improves milk quality and shelf life and makes it safer for consumers.

Only 30 litres of acaricides are used to spray 25 cows; prepare and use the recommended 75 litres of spray mixture for the 25 cows, following the product instructions as this ensures proper wetting of the animal with the acaricide killing ticks thus keeping the cattle healthier and allows them to use nutrients for growth and milk production.

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ITEM · 4

Part (a)

Over crowding (house of $1um^2$ for 300 layers); Over crowding causes competition for feed, water and space leading to stress, uneven feed intake, poor growth and increased disease transmission which cumulatively lower egg production.

Unlockable door; This allows entry of predators such as dogs and wild cats and thieves, causing loss of birds through predation and theft and also increases stress to the remaining flock, this reduces the number of birds available for production and disrupts laying.

Brooder not pre-warmed before chicks were introduced; chicks are unable to regulate their own body temperature at this age, so they suffer cold stress, huddle together and become weak resulting in high chick mortality and poor future layer stock

Unsexed chicks purchased contained 40% male; Male birds donot lay eggs yet they consume alot of feeds and occupy space meant for the layers wasting resources and reducing the actual number of eggs-producing birds in the flock resulting into low egg production.

Birds with fresh wounds around the vent and damaged combs; Poor wing feathering; These are signs of cannibalism caused by overcrowding and stress where injured birds lose blood, get infected and most of the energy, oxygen and nutrients that would go to egg production is instead used for healing.

Eggs collected only once every three days; Leaving eggs for so long exposes them to dirt, get cracked and broken and hens may start eating broken eggs giving rise to egg-eating habit that reduces the number of eggs that reach the market in good condition

Eggs found evenly distributed on the litter floor instead of in nest; Eggs layd on litter get contaminated with droppings and dirt and are more easily broken, lowering egg quality and the quantity fit for sale.

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4 Part. a

Only about 6 feeders / drinkers are provided for 300 chicks; This causes overcrowding at feeding points where weaker chicks are outcompeted and get insufficient feeds and water, leading to uneven growth, stunting and higher mortality which reduces the number of birds reaching laying stages.

Layers Mash fed to young chicks instead of chick mash; layer's mash has high calcium meant for eggshell formation in mature hens. In young chicks this damages the kidneys and does not meet their higher protein needs for growth, causing poor development and increased deaths.

Two layers found dead with twisted necks, some birds shivering and inactive; These signs indicate an outbreak of Newcastle disease, which spreads quickly through the flock, causing further deaths and a sharp drop in the number of birds laying eggs.

The house has a leaking roof; The leaking roof allows rainwater to enter the poultry house and wet the litter, the wet litter promotes the multiplication of disease-causing organisms and increases humidity and ammonia production, which can cause respiratory problems and poor bird health, ultimately reducing the productivity.

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ITEM - 4

Part (b)

The poultry house is too small for 300 layers; he should expand the poultry house or reduce the number of birds to match the available floor space, thus providing adequate space improves movement, feeding and ventilation giving rise to healthy, unstressed birds with better growth and egg production resulting into increased income.

Many birds are males although the farmer intended to keep layers; Separate and sell the unwanted males, retaining only the required males if the project is intended for breeding, this reduces feed and management costs and increases the proportion of productive females, therefore increasing on farmer's saleable eggs.

Birds have wounds around the vent and missing feathers; Reduce overcrowding and provide adequate feeder and balanced diet and treat injured birds. Proper space and nutrition reduces aggression and injuries while treating wounded birds prevents further complications and deaths, allowing more birds to remain productive.

Many eggs are being broken; Provide adequate laying nests, position them correctly and collect eggs frequently therefore when adequate nests are available, hens are more likely to lay eggs in protected locations rather than on the floor while frequent collection reduces the time eggs remain exposed to trampling. This reduces egg breakage and increases the number of eggs available for sale.

Eggs are being laid throughout the deep-litter house; Provide sufficient clean, comfortable and suitably positioned nests and encourage hens to use them as proper nests concentrate eggs in safe locations, reducing contamination, cracking and breakage thus increasing the proportion of marketable eggs.

Dead birds have twisted necks; Seek veterinary diagnosis immediately, isolate sick birds where appropriate and follow a recommended vaccination and disease-control programme as this helps to identify, treat and

4 Part. b

protect susceptible birds against preventable disease causing a reduction in mortality rate which preserves the productive flock and prevents economic losses.

Some birds are shivering and inactive; Maintain appropriate temperature and ventilation, prevent cold drafts and seek veterinary examination for persistent signs Therefore providing suitable environmental conditions allows more nutrients to be directed towards production improving productivity.

The poultry house has a leak-proof problem; Repair the roof immediately and keep the litter dry since wet litter increases humidity and encourages multiplication of disease-causing organisms therefore dry house improves bird health, reduces disease and allows better egg production..

The door is strong but not lockable; Install a proper lockable door and restrict unnecessary access to the poultry house, as a secure house prevents theft and unauthorized entry and reduces the introduction of disease organisms by the visitors hence protecting the flock and improving profitability.

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ITEM • 5

Part (a)

Hired land for only two years; Bananas are perennial crops that take time to establish and reach full production therefore a short hire period does not allow enough time to recoup establishment costs or benefits from high yields.

Upper part of the land has exposed roots and short trees with exposed roots This indicates that topsoil has been eroded away leaving poor, shallow soil with low fertility and provide weak anchorage for the suckers planted there, this makes them susceptible to slight changes in climate and low nutrient uptake resulting in stunted growth and reduced bunch size.

Soil PH is 2 : PH 2 is way too acidic and leads to unavailability of nutrients needed by the plants and also increases the prevalence of fungal diseases that attack the plants and reduce yield

Soil is dark, plastic and forms a ribbon when rolled and retains alot of water; clay soils have a poor drainage and aeration, abundance of water in the soil pores cuts off oxygen supply to the roots, this causes roots to respire anaerobic ally producing little ATP and ethanol responsible for root rot which reduces on nutrient uptake and water uptake leading to low yields.

Harvested bunches are piled under the sun for up to two days awaiting transport; Prolonged exposure to sunlight causes sunscald and accelerated over-ripening, leading to spoilage and post-harvest losses that reduce the amount of marketable produce.

Area is prone to banana weevils and banana wilt; Weevils bore into the corm and pseudostem weakening the plant and causing it to lodge while bacterial wilt destroys the vascular tissue, causing premature ripening and total plant death, hence resulting into reduced yields.

Prolonged dry spells and inadequate rainfall even during wet seasons; low rains makes the plants to suffer water stress and increased wilting, this causes plants to close up there stomata reducing surface area for CO_2 uptake for photosynthesis to occur, resulting into poor bunch filling and small yields.

5 Part. a

Insuppicient manure bought for 2,700 suckers; He brought 72,000 kg against 108,000 kg needed for suckers, the short pall of 36000kg means many plants will not receive adequate nutrients, resulting into stunted growth and low yields.

Few plant residues remain in the soil along the slope; This indicates low organic matter needed for decomposition into humus for increased portility, leading to poor soil structures and low nutrient reserves resulting to poor plant vigour and low yields.

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ITEM · 5

Part (b)

Soil PH of 2 ; Apply agricultural lime to raise the PH to the required 6-7 PH range, this will help to neutralize soil acidity, releasing bound nutrients and reducing the risk of fungal diseases thus improving nutrient uptake.

Hired land for only two years; Negotiate for a lonyer-term lease agreement with land owner because bananas need several years to reach peak, sustained production, this will allow full returns on establishment costs.

Few plant residues along the slope; incorporate crop residues and mulch on the land as this will improve soil structure, water retention and nutrient cycling thus boosting fertility.

Area is prone to banana weevils and bacterial wilt; Use clean disease free suckers, practice field sanitation, disinfect tools between plants and control wee vils using appropriate insecticides, this prevents introduction and spread of pests and bacterial wilt, protecting plant health and yield.

Insuffeicient manure ; lurchase additional manure to meet the full plant requirement or supplement with inorganic fertilizer, this ensures that every plant receives adequate nutrients for optimum growth and yield.

Bananas left piled in the sun after harvest ; Arrange transport in advance and provide shade for harvested bunches. this prevents sunscald, over-ripening and spoilage, preserving quality and marketable yield.

Upper part of the land has exposed roots and short trees with exposed roots; Practice soil and water conservation and add topsoil manure to affected areas this reduces further soil loss and restores fertility, improving root anchorage and nutrient supply.

Prolonged dry spells and inadequate rain falls; Install irrigation schemes especially drip irrigation method as this concentrates water to the roots directly, thus reaching the 1,200 - 2,500 mm /year water required for banana growth and prevents water stress, leading to higher yields.

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5 part . b

Heavy soils that retain alot of water; Dig drainage channels to improve drainage and always plant on raised ridges, this helps to remove excess water from the root zone ensuring aeration of the soil needed for aerobic respiration to obtain ATP for nutrient uptake.

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ITEM - 6

Part (a)

Obtained one hectare on communal land with agreement by one member; This may not provide secure and recognized rights over the land hence discouraging long-term investments in soil improvements and conservation leading to the declining soil fertility and erosion, resulting into low bean yields

The field had been over cultivated; continuous cultivation depletes soil nutrients such as nitrogen and phosphorus without allowing time for the restoration and encourages build-up of particular pests, diseases and weeds which cumulatively caused damage to the crops, resulting into low yields.

Plant residues were usually burnt before planting; Burning destroyed the organic matter reducing there biomass and heat from burning killed the useful micro organisms that cause decomposition and nutrient cycling, this makes the soil infertile resulting into low yields.

Applied 4 litters of foliar fertilizer per hectare; Excessive fertilizer application increased the hypertonic conditions around the leaves causing efflux of water by exosmosis from the plant guard cells, this makes leaves flaccid and drop interrupting carbon dioxide uptake needed for photosynthesis hence the low yields.

Weeds were removed late from the garden; Delayed weeding application enabled weeds to accumulate in the garden, allowing them to out compete the bean plants for nutrients, water, light and space, This reduced crop growth leading to low crop yields.

Delayed control measures after evident signs of pest damage; Delayed control allowed the pests population to increase and cause cumulative damage to leaves, stems, flowers and pods, this reduced photosynthesis and pollination efficiency resulting into low yields.

No clear schedule existed for field operations; Since planting, weeding, Pest control and harvesting are time sensitive, lack of a schedule made

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6 part . a

These operations to be performed late allowing problems such as weeds, pests to reach damaging levels before action is taken hence leading to low yields.

Seeds were planted at varying depth; uneven planting depth causes uneven germination since deeply planted seeds receive little moisture, air and temperature and may use up their stored food reserves before reaching the soil surface, this results in complete failure to emerge which reduces plant population leading to low yields

Soil appeared lose and prone to soil erosion; lose soils are easily carried away by erosive agents such as flowing water and wind, this removes the nutrient-rich upper layer of soil and also provides weak anchorage to plant roots hence plants remain on top and receive few nutrients for their growth leading to low yields.

Some pods shattered before collection during harvesting time; Mature pods broke open and the seeds were lost in the field, causing hiring of larger labour force to pick scattered seeds hence increasing on production costs and direct reduction in quantity of harvested seeds leading to low profits.

Some plants showed poor growth and pale leaves; Pale leaves indicated the nutrient deficiency risk specifically nitrogen needed for chlorophyll formation and protein formation, this reduces on the light trapping efficiency of pale leaves for photosynthesis to occur thus resulting into weak and stunted plants which give low yields

ITEM - 6

Part (b)

Obtained one hectare on communal land with agreement by one member; Always obtain a clear, recognized agreement from the authority managing the communal land preferably in a documented form. this gives the farmer greater security of use and encourages them to invest in soil improvement and long-term practices that sustain high yield production.

The field had been over cultivated: Practice crop rotation and avoid continuous cultivation of beans on the same land to allow time for restoration of the depleted nutrients while crop rotation interrupts pests and disease cycles hence leading to soil fertility

Weeds were removed late allowing them to out compete crops; Carryout early and regular weeding especially during the early growth stages to prevent weeds from competing with beans for water, nutrients, light and space to allow rapid crop growth to maturity stages

Delayed pest control measures after evident signs of pest damage; Regularly scout the field and control pests promptly using appropriate intergrated pest-management methods, as early detection prevents pest population from reaching damaging levels thereby protecting leaves, flowers and pods responsible for yield formation.

No clear schedule existed for field operations; Always prepare and follow a farm calender for planting, weeding, pest control and harvesting, timely operation ensures that management practices are performed when the crops need them most hence reducing avoidable yield losses.

Seeds were planted at varying depth; Plant seeds uniformly at the recommended depth of 4 - 5 cm deep, uniform depth provides relatively similar moisture, oxygen and temperature conditions required for the germination and emergence of the shoots and roots

Applied 4 littres of foliar fertilizer per hectare; Apply the recommended 0.8l amount of fertilizer per hectare to prevent fertilizer injury to the plant

while supplying nutrients at a concentration that the crops can utilize effectively

Plant residues were usually burnt before planting; incorporate residues into the soil instead of burning, this adds organic matter and also improves soil fertility, structure and water-holding capacity needed for normal plant growth.

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